Complex variable theory – complex functions and residues

Complex numbers were invented in the sixteenth century to solve cubic equations and were long regarded as “imaginary”. Today they are indispensable in physics and engineering: alternating current, quantum mechanics, signal processing and the flow round a wing are all calculated with complex functions. Complex Variable Theory shows that a complex-differentiable function has remarkably rigid properties, and that integrals can be evaluated simply by counting poles. The page has an interactive colouring of complex functions and a figure on contour integrals.

> University of Sussex, Complex Variable Theory (1975–78): One of the mathematics courses in the physics syllabus. The page describes the subject as it is normally taught in a physics degree; it is not a reproduction of the original course.

Complex numbers

A complex number $z = x + iy$ is a point in the plane. Multiplying by i is a rotation by 90°, and in general lengths multiply and angles add: in polar form $z = re^{i\theta}$, $z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$. The link is Euler’s formula

$$e^{i\theta} = \cos\theta + i\sin\theta, \qquad e^{i\pi} + 1 = 0$$

which unites the exponential function with the trigonometric functions. That is why oscillations are so easy to handle with complex numbers: $e^{i\omega t}$ is a vector rotating with angular velocity ω, and its real part is a cosine oscillation (see the phasors in circuit theory).

Analytic functions

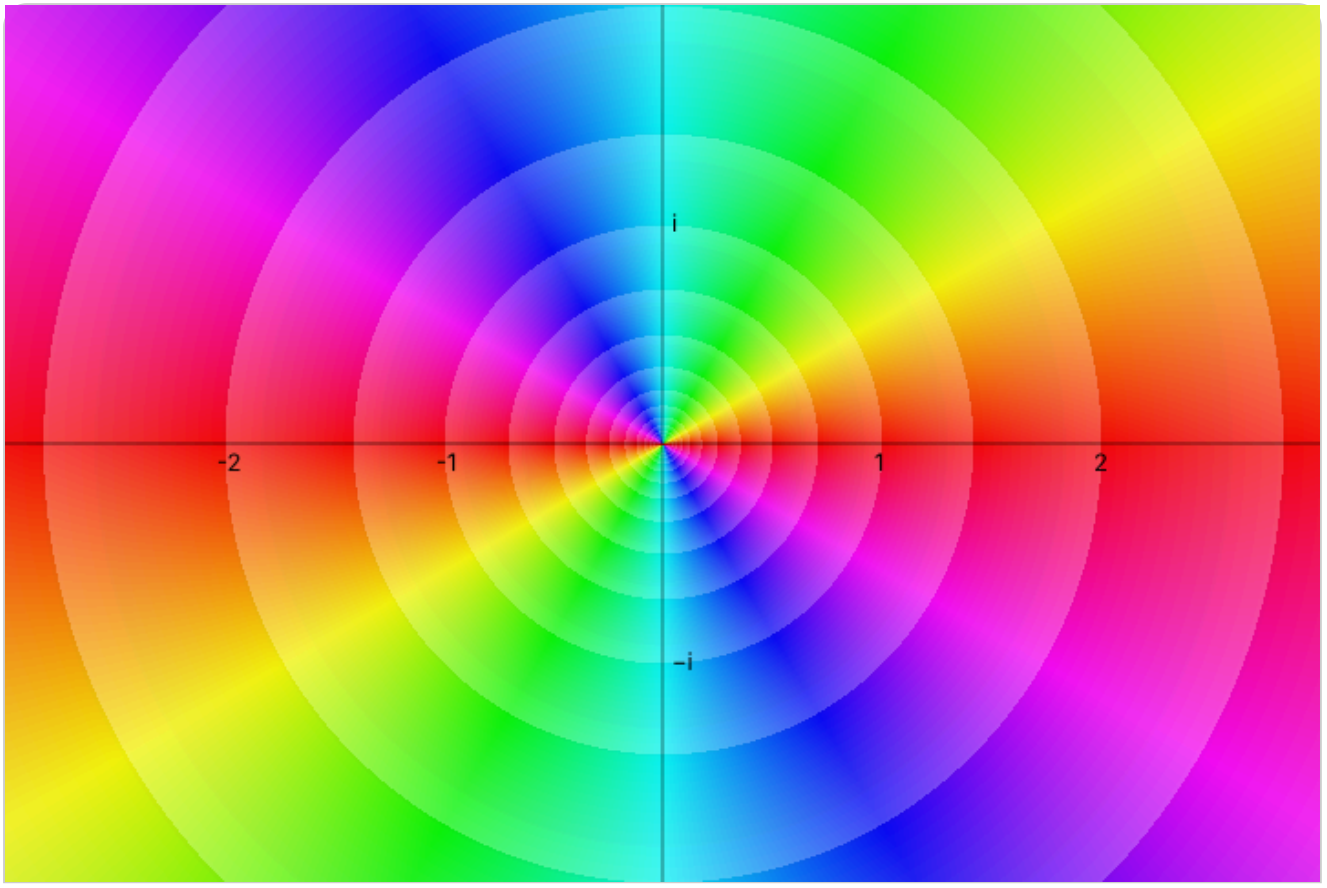
A complex function $f(z) = u(x,y) + iv(x,y)$ is differentiable if the limit $f'(z) = \lim_{h\to 0}(f(z+h) - f(z))/h$ is the same whatever direction h approaches zero from. That is a strong requirement, and it is equivalent to the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

A function that is differentiable in a region is called analytic (holomorphic). It is then automatically differentiable infinitely many times and equal to its Taylor series. Both u and v satisfy Laplace's equation $\nabla^2 u = 0$ – which is why analytic functions can be used to solve two-dimensional problems in electrostatics, heat conduction and flow.

Functions in colour

A complex function maps a plane onto a plane and cannot be drawn as an ordinary graph – that would need four dimensions. An elegant solution is *domain colouring*: each point z is coloured according to the value $f(z)$, with the hue given by the argument (angle) and the brightness by the magnitude.



Conformal maps

An analytic function preserves angles: small figures are rotated and scaled, but not distorted. That makes it possible to map a difficult region onto an easy one. The classic example is Nikolai Zhukovsky's map $w = z + 1/z$ from 1910, which maps a circle onto an aerofoil. The flow round a cylinder is known exactly, and the map gives the flow round the wing at once – and the lift via the Kutta–Zhukovsky theorem $L = \rho U \Gamma$ (see the wind tunnel and fluid mechanics).

Cauchy's theorem

Augustin-Louis Cauchy showed around 1825 that the integral of an analytic function round a closed curve is zero if the function is analytic everywhere inside the curve:

$$\oint_C f(z)\,dz = 0, \qquad f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}\,dz$$

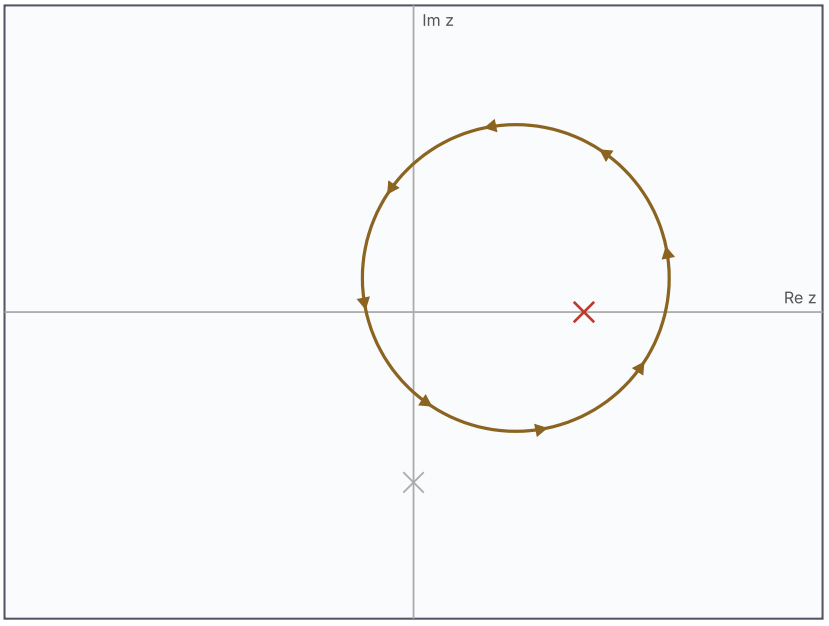
The second formula, Cauchy's integral formula, is even more surprising: the value of f at an interior point is completely determined by its values on the boundary. It is the complex version of the mean-value property of harmonic functions (see applied field theory).

Residues

If f has isolated singularities, it can be expanded near each of them in a Laurent series with negative powers, $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$. The coefficient a_{-1} is the *residue*, and the residue theorem says that

$$\oint_C f(z)\,dz = 2\pi i \sum_{k} \operatorname{Res}(f, z_k)$$

summed over the singularities inside the curve. For a simple pole $\operatorname{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z)$. With the theorem many real integrals can be evaluated that are otherwise impossible – for example $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$, which comes from the residue at $z = i$ – and in physics it is used to compute Fourier transforms, Green's functions and the Kramers–Kronig relations between refractive index and absorption.



Applications

- **Electrical engineering:** Impedances, transfer functions and stability. An amplifier with feedback is stable if the poles of its transfer function lie in the left half-plane – Nyquist's criterion (1932) is built directly on the argument principle.
- **Flow and electrostatics:** Two-dimensional potential problems are solved with complex potentials and conformal maps.
- **Quantum mechanics:** The wave function is complex, and poles of scattering amplitudes in the complex energy plane correspond to bound states and resonances.
- **Acoustics:** Sound propagation in a pipe with an absorbing lining has complex propagation constants, and the roots of the transcendental equation are found in the complex plane – exactly the problem in my thesis at DTH.

Since 1978

- **Fractals:** The Mandelbrot set (1980) – the values of c for which the iteration $z \mapsto z^2 + c$ stays bounded – made complex functions one of the best-known images of the age.
- **Computer graphics:** Domain colouring only became practical with colour screens and is now the preferred way to show complex functions.
- **The Riemann hypothesis** on the zeros of the zeta function $\zeta(s)$ is still unproved and one of the Clay Institute's seven Millennium Prize Problems from 2000.

Further reading

- Tristan Needham: *Visual Complex Analysis* (25th anniversary ed., Oxford 2023)
- Ruel V. Churchill and James Ward Brown: *Complex Variables and Applications* (9th ed., McGraw-Hill 2013)
- Elias Wegert: *Visual Complex Functions. An Introduction with Phase Portraits* (Birkhäuser 2012)